RAMANUJAN'S CONSTANT AND THE ZETA FUNCTION ON ONE PAGE

The Basel sum hidden in $e^{\pi\sqrt{163}}$

Writing π through $\zeta(2)$ turns Ramanujan's near-integer into thirty digits of the Basel sum.

THE RULE

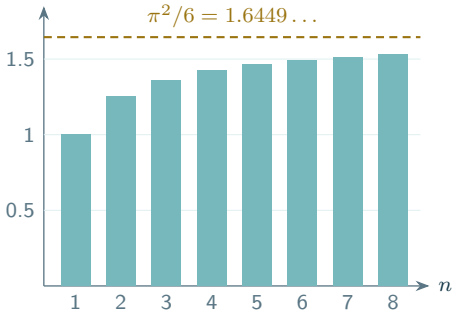
Replace π by the value of a zeta function at 2

Euler's solution of the Basel problem, and Riemann's completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ evaluated at the same point, give

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}, \qquad \xi(2) = \frac{\pi}{6},$$

so $\pi = \sqrt{6\,\zeta(2)} = 6\,\xi(2)$. Substituting into Ramanujan's constant moves $\zeta(2)$ inside the square root, or brings $\xi(2)$ out of it entirely:

$$\mathcal{R} \;=\; e^{\pi\sqrt{163}} \;=\; e^{\sqrt{978\,\zeta(2)}} \;=\; e^{6\sqrt{163}\,\xi(2)}$$



Partial sums of $\sum 1/n^2$ creep up towards Euler's $\pi^2/6$.

THE SHORT FORM OF RAMANUJAN'S CONSTANT

Thirty digits of π and of $\zeta(2)$ from one logarithm

Since $\mathcal{R}$ falls short of $640320^3 + 744$ by only 7.5×10^{-13}, a relative error of 3×10^{-30}, taking logarithms of the short form $\mathcal{R} \approx 640320^3 + 744$ loses almost nothing:

$$\zeta(2) \;\approx\; \frac{\left[\ln(640320^3+744)\right]^2}{978}, \qquad \xi(2) \;\approx\; \frac{\ln(640320^3+744)}{6\sqrt{163}}, \qquad \pi \;\approx\; \frac{\ln(640320^3+744)}{\sqrt{163}}$$

short form	$3.141\,592\,653\,589\,793\,238\,462\,643\,383\,279\,726\ldots$
true π	$3.141\,592\,653\,589\,793\,238\,462\,643\,383\,279\,502\ldots$

The two agree to thirty decimal places, and the expression for $\zeta(2)$ matches $1.644\,934\,066\,848\ldots$ just as far. The integer 640320^3 is $-j\left(\frac{1+i\sqrt{163}}{2}\right)$, so the digits of the Basel sum are being read off a value of the j-function.

FROM SQUARES TO PRIMES

Euler's product puts the primes inside ζ

For $s > 1$, unique factorisation turns the sum into a product over primes,

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}, \qquad \text{so} \qquad \frac{\pi^2}{6} = \frac{4}{3}\cdot\frac{9}{8}\cdot\frac{25}{24}\cdot\frac{49}{48}\cdots,$$

where primes up to 29 already give 1.633, and $6/\pi^2 \approx 0.608$ is the chance that two random whole numbers share no prime factor.

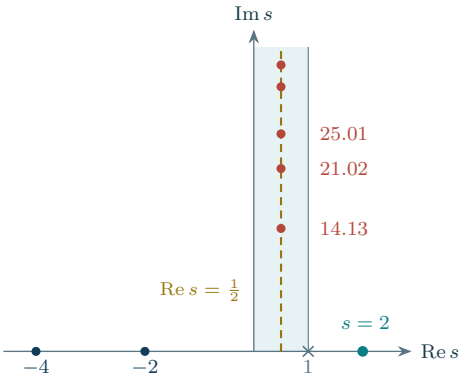
THE RIEMANN HYPOTHESIS

The zeros of ξ all lie on one line

Riemann extended $\zeta(s)$ to complex s in 1859; it vanishes at the trivial zeros $-2, -4, \ldots$ and otherwise only in the strip $0 < \operatorname{Re} s < 1$. The completed ξ, the function that gave $\xi(2) = \pi/6$, cancels the pole and the trivial zeros and satisfies $\xi(s) = \xi(1-s)$, so its zeros are exactly the non-trivial zeros of ζ.

Riemann hypothesis: every zero of $\xi(s)$ has real part $\frac{1}{2}$.

The first zeros sit at heights 14.1347, 21.0220 and 25.0109, and more than 10^{13} have been checked. If it holds, the count of primes up to x differs from $\int_2^x \frac{dt}{\ln t}$ by at most about $\sqrt{x}\,\ln x$.



Trivial zeros, the pole at 1, Basel's $s = 2$ and the first zeros on the critical line.

Back to 163. The generalised hypothesis concerns the zeta functions of fields such as $\mathbb{Q}(\sqrt{-163})$; Hecke, Deuring and Heilbronn showed that the class numbers of $\mathbb{Q}(\sqrt{-d})$ grow without limit whether or not it is true, which is why the list of Heegner numbers, and of near-integers like $\mathcal{R}$, had to be finite. Here Γ is the gamma function, with $\Gamma(1) = 1$.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames